

Build Knowledge Graph RAG With LlamaIndex for Data Analysis

Project Overview

Overview

The financial market is a complex and dynamic ecosystem that generates vast amounts of data every second, including stock prices, news updates, industry trends, and economic reports. Analyzing this data is critical for investors and analysts to make informed decisions and manage risks. Traditionally, market data analysis has relied on manual efforts or basic tools that extract and process data from individual sources. This approach, while effective to some extent, is time-consuming, prone to errors, and limited in uncovering deeper insights such as hidden patterns or relationships between entities like companies, sectors, and stocks.

Generative AI (Gen AI) offers transformative potential in market data analysis by automating data extraction, processing, and structuring. With its ability to process unstructured data, Gen AI can uncover complex relationships and provide real-time, actionable insights. By combining web scraping, natural language processing (NLP), and knowledge graph construction, AI systems can not only handle the scale and speed required in financial markets but also offer personalized, context-aware insights to users. This allows professionals to quickly navigate the overwhelming sea of market data and focus on strategic decision-making.

In this project, we leverage the capabilities of Gen AI to create a Knowledge Graph RAG system for market sentiment data analysis. Using web scraping techniques, we gather real-time data from multiple sources such as stock feeds, financial news websites, and industry reports. This structured information is used to build a knowledge graph database with Neo4j, LlamaIndex and OpenAI. We then query the graph index for generating context-aware responses to user queries.

Prerequisite Project: Kindly ensure the completion of the [LLM Project for building and fine-tuning a large language model](#) before proceeding with this project.

Prerequisites: Python programming knowledge, environment setup (virtual environments, dependency management), prompt engineering, understanding and using large language models (LLMs)

Aim

The aim of this project is to do financial market sentiment analysis by leveraging AI to automate data extraction, structure insights into a knowledge graph, and provide actionable insights for informed financial decision-making.

Data Description

The project utilizes diverse financial data from real-time stock feeds, news articles, and market reports. This unstructured data is processed to extract key entities and relationships, which are then structured into a dynamic knowledge graph for analysis.

Tech Stack

- **Language:** Python 3.10
- **Libraries:** pandas, BeautifulSoup, requests, neo4j, LlamaIndex, openai
- **Models:** OpenAI GPT 4o

Approach

- **Data Collection:** Gather financial data from multiple sources such as stock feeds, news websites, and industry reports using Google search engine API and scraping techniques
- **Knowledge Graph Construction:** Build a knowledge graph to represent relationships between extracted entities like companies, market trends, and sectors using neo4j
- **Vector Embedding and Indexing**
 - Convert textual data into vector embeddings using OpenAI
 - Index data using LlamaIndex to enable retrieval-augmented generation
- **RAG Workflow Integration**
 - Query the vector database to fetch relevant context and related entities from the knowledge graph
 - Use OpenAI to generate contextually accurate and actionable responses

Modular code overview:

Once you unzip the modular_code.zip file, you can find the following:

```
graph-rag
├─ dataset.zip
├─ engine.py
├─ notebook
│   └─ updated_market_data_analysis.ipynb
├─ readme.md
├─ references
│   └─ Solution Methodology.pdf
├─ requirements.txt
├─ src
│   ├── constants.py
│   ├── graph_store_creation.py
│   └─ utils.py
└─ storage
    ├── default__vector_store.json
    ├── docstore.json
    ├── graph_store.json
    ├── image__vector_store.json
    └─ index_store.json
```

Here is a brief information on the files:

- **Root Directory:** Contains the main files like `dataset.zip` (data), `engine.py` (core logic), `readme.md` (project overview), and `requirements.txt` (dependencies).
- **Notebook:** Includes `updated_market_data_analysis.ipynb` for analysis and insights.
- **References:** Holds supporting documents, such as `Solution Methodology.pdf`.
- **Source Code (src):** Includes scripts for constants (`constants.py`), graph store creation (`graph_store_creation.py`), and utilities (`utils.py`).
- **Storage:** Stores various JSON files for vector stores, document storage, and graph storage configurations (e.g., `default__vector_store.json`, `graph_store.json`).
- **Kindly follow all the instructions for running the code from Readme.md file.**

Project Takeaways

1. Understand how to automate data collection from diverse financial sources like stock feeds, news websites, and industry reports using Google Search API and web scraping techniques
2. Learn how to build and structure a knowledge graph using Neo4j to represent entities (e.g., companies, sectors, trends) and their relationships
3. Gain hands-on experience in generating vector embeddings for textual data using OpenAI and learn how they capture semantic meaning
4. Discover how to index and manage embedded data efficiently using LlamaIndex for optimized retrieval
5. Learn the principles of Retrieval-Augmented Generation (RAG) and how it enhances AI responses by combining retrieved data with generative models
6. Explore the integration of knowledge graphs into a RAG workflow to provide deeper, contextually rich, and actionable insights
7. Understand how to use cutting-edge tools like Neo4j, OpenAI, and LlamaIndex for data analysis